JAW CRUSHER

(MO-101)

Foreword

Welcome to the fast growing family of K.C. product owners. We appreciate your interest in us and thank you for buying our product.

You have chosen the finest quality product in the market which is produced using latest techniques and has underwent strict quality control tests. It is a product that we are proud to build and you are proud to own it.

Our products are easy to understand and operate. They are excellent for students who are trying to gain practical knowledge through experiments.

However your comfort and safety are important to us, so we want you have an understanding of proper procedure to use the equipment. For the purpose, we urge you to read and follow the step-by-step operating instructions and safety precautions in this manual. It will ensure that your favourite product delivers reliable, superior performance year after year.

This manual includes information for all options available on this model. Therefore, you may find some information that does not apply to your equipment.

All information, specifications and illustrations in this manual are those in effect at the time of printing. We reserve the right to change specifications or design at any time without notice.

Customer satisfaction is our primary concern. Feel Free to contact us for any assistance. So what are you waiting for, roll up your sleeves and let us get down to work!

K.C. Engineers Pvt. Ltd.

Important Information About This Manual

Reminder for Safety

Modification on Equipment:

This equipment should not be modified. Modification could affect its performance, safety or disturbance. In addition damage or performance problems resulting from modification may not be covered under warranties.

Precautions and Maintenance:

This is used to indicate the presence of a hazard that could cause minor or moderate personal injury or damage to your equipment. To avoid or reduce the risk, the procedures must be followed carefully.

Contents

JAW CRUSHER

1. OBJECTIVE:

To study the operation of jaw crusher.

2. AIM:

To determine the efficiency of the crusher for crushing a material of known work index (W_i).

3. INTRODUCTION:

Jaw crushers do the heavy work of breaking large pieces of solid material into small lumps. The jaw crusher is widely used in industry for coarse reduction of large quantities of solids. They operate by compression and can break large lumps of very hard materials, as in the primary and secondary reduction of rocks and ores. They are very common in industry and have a wide application.

4. THEORY:

In a jaw crusher feed of known size distribution is admitted between the two jaws, set to form a V open at the top. It is driven by an eccentric so that a great compressive force is applied to lumps of solids caught between the jaws. Large lumps caught between the upper parts of the jaw are broken, drop into narrower space below and are re-crushed the next time the jaws close. The most common type of jaw crusher is the blake crusher. In this machine an eccentric drives a pitman connected to two toggles, one of which is pinned to the frame and the other to the swinging jaw. The pivot point is at the top of the movable jaw or above the top of the jaws on the centerline of the jaw opening. The greatest amount of motion is at the bottom of the V, which means that there is little tendency for a crusher of this kind to choke.

BOND CRUSHING LAW AND WORK INDEX:

A more realistic method of estimating the power required for crushing and grinding is

$$\frac{P}{m} = \frac{K_b}{\sqrt{D_p}} \qquad \text{---------------- (1)}$$

Where K_b is a constant which depends on the type of machine and on the material being crushed, D_p is particle size in millimeters, P is power in kilowatts and m is mass flow rate in tons per hour.

W_i is defined as the gross energy requirements in kilo watt hours per ton of feed

needed to reduce a very large feed. This definition leads to a relation between K_b and W_i.

$$K_b = 0.3162 * W_i \qquad \text{--------------- (2)}$$

If 80 percent of the feed passes a mesh size of D_{pa} mm and 80 percent of the product a mesh of D_{pb} mm, it follows from eq (1) & (2).

$$\frac{P}{m} = 0.3162 \times W_i \left(\frac{1}{\sqrt{D_{Pb}}} - \frac{1}{\sqrt{D_{Pa}}} \right)$$

$$P = m \times 0.3162 \times W_i \left(\frac{1}{\sqrt{D_{Pb}}} - \frac{1}{\sqrt{D_{Pa}}} \right) \qquad \text{--------------- (3)}$$

5. DESCRIPTION:

The set-up is a blake jaw crusher, contains two jaws of hard steel with one jaw stationary and other is moving. A hopper is provided at the top for feeding material. The opening of the jaw is adjustable. The motor is coupled to the machine through triple `V' belt drive. A handle is provided.

6. UTILITIES REQUIRED:

6.1 Electricity Supply: Single Phase, 220 V AC, 50 Hz, 5-15 Amp combined socket with earth connection

6.2 Floor Area Required: 1m x 1m.

6.3 Raw material for feed (max size 50 mm)

6.4 Set of sieves with sieve shaker for analysis.

7. EXPERIMENTAL PROCEDURE:

7.1 STARTING PROCEDURE:

7.1.1 Prepare a suitable feedstock of a solid material.

7.1.2 Measure its size distribution.

7.1.3 Fix the opening of jaws.

7.1.4 Switch ON the power supply.

7.1.5 Start the machine with no load condition and record the time taken for 10-20 pulses of energy meter.

7.1.6 Start feeding the solid material by hopper at a constant rate.

7.1.7 Again record the time taken for 10-20 pulses of energy meter.

7.1.8 Repeat the experiment for different opening of jaws.

7.2 Closing Procedure:

7.2.1 When experiment is over switch OFF the power supply.

8. Observation & Calculation:

8.1 Data:

Energy meter constant EMC = 3200 Pulses/kWh

Work Indexes Of Some Common Minerals

Material	Work Index (W_i)
Bauxite (sp.gr =2.20)	8.78
Cement climker (sp.gr =3.15)	13.45
Coal (sp.gr =1.40)	13.00
Coke (sp.gr =1.31)	15.13
Gravel (sp.gr =2.66)	16.06
Gypsum rock (sp.gr =2.69)	6.73
Lime stone (sp.gr =2.66)	12.74
Quartz (sp.gr =2.65)	13.57

Observations:

W_f = _____ kg

t_c = _____ sec

D_{pa} = _____ mm

D_{pb} = ______ mm

8.2 Observation Table:

P_1	t_{p1} (sec)	P_2	t_{p2} (sec)

8.3 Calculations:

$$P_{NL} = \frac{P_1 \times 3600}{t_{p1} \times EMC} \text{ (kW)}$$

$$P_L = \frac{P_2 \times 3600}{t_{p2} \times EMC} \text{ (kW)}$$

$$P_{act} = P_L - P_{NL} \text{ (kW)}$$

$$m = \frac{W_f}{t_c} \times \frac{3600}{1000} \text{ (tons/h)}$$

$$K_b = 0.3162 \times W_i \text{ (kWh/tons)}$$

$$P_{cal} = m \times K_b \times \left[\frac{1}{\sqrt{D_{Pb}}} - \frac{1}{\sqrt{D_{Pa}}} \right] \text{ (kW)}$$

$$\eta = \frac{P_{act}}{P_{cal}} \times 100 \text{ (\%)}$$

9. Nomenclature:

Nom	Column Heading	Units	Type
D_{pa}	Average feed size	mm	Measured
D_{pb}	Average product size	mm	Measured
EMC	Energy meter constant	Pulses/kWh	Given
K_b	Bond's constant	kWh/tons	Calculated
m	Feed rate	tons/ h	Calculated
P_1	Number of pulses counted at no load condition	*	Measured

P_2	Number of pulses counted at loaded condition	*	Measured
P_{act}	Actual power required for crushing	kW	Calculated
P_{cal}	Calculated power required for crushing	kW	Calculated
P_L	Power consume by machine at loaded condition	kW	Calculated
P_{NL}	Power consume by machine at no load condition	kW	Calculated
t_c	Time for crushing	sec	Measured
t_{p1}	Time for P_1 pulses	sec	Measured
t_{p2}	Time for P_2 pulses	sec	Measured
W_f	Weight of feed taken	kg	Measured
W_i	Work index of material	kWh/tons	Given
η	Crushing efficiency	%	Calculated

* Symbols are unit less.

10. PRECAUTION & MAINTENANCE INSTRUCTIONS:

10.1 Never run the apparatus if power supply is less than 180 volts and more than 230 volts.

11. TROUBLESHOOTING:

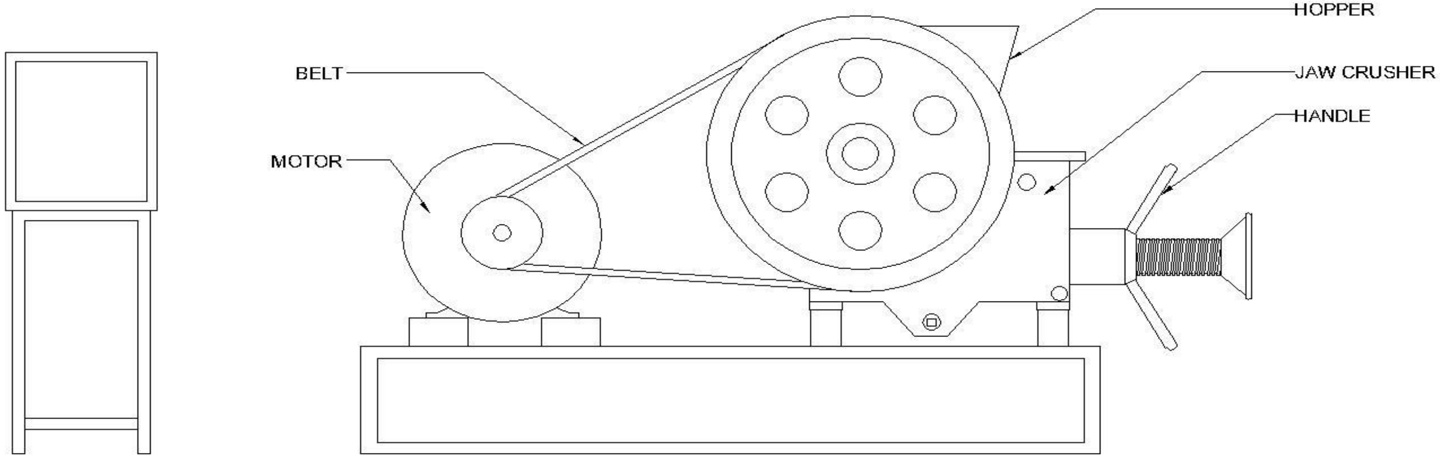
11.1 If electric panel is not showing the input on the mains light, check the main supply.

12. REFERENCES:

12.1 McCabe, Warren L. Smith, Julian C. Harriott, Peter (2005). *Unit Operations of Chemical Engineering.* 7th Ed. NY: McGraw-Hill. pp 985-986, 988.

12.2 Brown, George Granger (1995). *Unit Operations.* 1st Ed. ND: CBS Publishers & Distributors. pp 27-28.

13. BLOCK DIAGRAM:



CONTROL PANEL

FRONT VIEW

BLOCK DIAGRAM

NAME - JAW CRUSHER

CODE - MO-101

VALVES

N / A